

TIA (U4 LMP7721): $V_{out} = V_{REF} \pm I_{det} \times R_f$
 $R_f = 2M\Omega \rightarrow 1.14 \mu A$ full scale (CEM), $57 \mu A / LSB$
 $R_f = 100M\Omega \rightarrow 25 nA$ full scale (Faraday), $1.25 pA / LSB$
 C1 sets the TIA pole: 7.2 kHz on 2M Ω , 159 Hz on 100M Ω
 R3/C2 = 1k Ω /1uF anti-alias: $f_c = 159$ Hz (Nyquist at 860 SPS is 430 Hz)
 D1 clamps DET_IN to $V_{REF} \pm 0.6 V$ at zero bias



I2C uses the dedicated SDA/SCL pins only: that works on UNO R3/R4, Leonardo and Mega. A4/A5 stay free.



$VREF = 5V \times R8 / (R7 + R8) = 2.50 V$. C3 filters USB ripple. U3B buffers it; R15 lets it drive C9. Feeds R17 $\rightarrow$ TIA IN+, the guard, D1 and AIN1.

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ADDR -> GND sets the I2C address to 0x48. C5 decouples U2

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A0 -> GND sets the I2C address to 0x60

U3A buffers the DAC (gain = 1) -> 0-5 V mass command. R12 isolates the cable.
A 0-10 V Extrel command needs a separate >=12 V rail
for U3 and gain 2 - NOT possible on the 5 V shield rail.

U3 is a dual MCP6002. A buffers the DAC. B buffers VREF. C4/C5/C6/C8 sit at the VDD pins of U1/U2/U3/U4. C7 at the 5 V output.

U3 is a dual MCP6002. A buffers the DAC, B buffers VREF.
C4/C5/C6/C8 sit at the VDD pins of U1/U2/U3/U4,
C7 at the 5 V entry.

2. Signal is a shield for an Arno UNO R3. All power comes from the UNO 5 V rail; no separate supply is fitted.
3. Binary chain: detector $\rightarrow$ J1 $\rightarrow$ R4 + D1 clamp $\rightarrow$ U3B TIA $\rightarrow$ R3/C2 anti-alias $\rightarrow$ U2 AINO $\rightarrow$ I2C $\rightarrow$ U5B $\rightarrow$ USB.
4. The ADS1115 is read single-ended at AINO with AIN1 tied to VREF, so AINO-AIN1 can also be read differentially.
5. The I2C bus is terminated at the Arno UNO R3 with a 100 nF capacitor. The I2C bus pull-up is 10 Ω ; tied to GND/V5.
6. GND/V5 each divide would leak 20–50 pA straight into the measured current. Do not substitute a switching diode.
7. ALERT/RDY is open-drain. R11 pulls it up; R10 can be used as a conversion-ready interrupt instead of polling.
8. Triger: Team 4's sequencer drives $\sim$ 1.1. F10 keeps DB defined when the cable is absent.
9. The cable shield is connected to GND/V5 common. Its own HV supply and its own cable shield is this board.
10. ADS115 inputs must stay inside GND,VDD. VREF at mid-supply gives 2.5 V of headroom either way, so the board reads a detector that sources current (Faraday cup) or one that sinks it (CD-SEM anode) unchanged.
11. The cable shield is connected to GND/V5 common. Its own HV supply and its own cable shield is this board. $C = 100$ pF, $R = 100$ Ω , $f = 1.7$ MHz and $Rf = 2$ M Ω that is 0.65 pF. $C_1 = 10$ pF gives the 4.32 kHz (2M Ω) / 159 Hz (100M) pole with a wide margin.
12. R3/C2 = 1k/15F: 159 Hz anti-alias corner, below the 720 kHz Nyquist at 860 SPS. 1k rather than 10k keeps the ADS1115 input current ($\sim$ 0.4 uA) from making a 4 mV offset. Set settling $\sim$ 11 ms, $\sim$ 30 s per 2600-point count.
13. The cable shield is connected to GND/V5 common. Its own HV supply and its own cable shield is this board.
14. MCP4725 powers up at mid-scale (2.5 V on the QMS). Program its EEPROM to 0X000 one from firmware.
15. The input sits at VREF = 2.5 V while the chamber and cable shield are at GND; insulation leakage is 2.5 pA per 10 pOhm. The cable shield is connected to GND/V5 common. Its own HV supply and its own cable shield is this board.
16. AIN2 reads the delivered QPS voltage back, so firmware can correct for the 5 V USB rail (the DAC's reference).

J1 BNC centre = detector / preamp output, shell = board GND. HV dynode bias never touches this board
 J2 1 = QPS_OUT 0-5 V mass command (via R12 100R), read back on AIN2 2 = GND
 J3 1 = TRIG IN, Team 4 sequencer, 0-5 V logic -> UNO D8 2 = GND
 J4 1 = VREF (2.50 V) 2 = AIN3 (spare ADC input) 3 = GND

JP1 1-2 = CEM range $R_f = R1\ 2M2$ full scale 1.14 uA, 57 pA/LSB
 2-3 = Faraday range $R_f = R2\ 100M$ full scale 25 nA, 1.25 pA/LSB
 I2C ADS1115 = 0x48 (ADDR->GND) MCP4725 = 0x60 (A0->GND)
 UNO D5 = ALERT/RDY D8 = trigger SDA/SCL pins = I2C. Every other UNO pin is free.

Firmware base: Grinits et al., J. Chem. Educ. 2016, 93, 1316
 Digi-Key and LCSC (JLPCPB assembly) part numbers in BOM.csv
 Team 5 scope: ADC side, MCP4725 mass command is fitted for Team 4's sequencer.
 Portable quadrupole mass spectrometer – data acquisition and control block
MI1001 Team 5 – Foundation of Imaging Engineering, University College Maastricht
 Sheet: /
 File: mie1001_daq_board.kicad_sch

Title: DAQ shield for Arduino UNO R3 – Team 5 acquisition board

Size: A3	Date: 2026-09-22
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